

*MA.7.NSO.1.2***Benchmark**

MA.7.NSO.1.2 Rewrite rational numbers in different but equivalent forms including fractions, mixed numbers, repeating decimals and percentages to solve mathematical and real-world problems.

Example: Justin is solving a problem where he computes $17/3$ and his calculator gives him the answer 5.6666666667. Justin makes the statement that $17/3=5.6666666667$; is he correct?

Connecting Benchmarks/Horizontal Alignment

- MA.7.AR.1.2
- MA.7.AR.3.1, MA.7.AR.3.3
- MA.7.DP.1.4
- MA.7.DP.2.2, MA.7.DP.2.4

Terms from the K-12 Glossary

Rational Number

Vertical Alignment**Previous Benchmarks**

- MA.6.NSO.1.1, MA.6.NSO.1.2
- MA.6.NSO.3.5

Next Benchmarks

- MA.8.NSO.1.1, MA.8.NSO.1.2

Purpose and Instructional Strategies

In grade 6, students rewrote positive rational numbers in different but equivalent forms as long as the decimal form is terminating. This expectation expands to all rational numbers in grade 7, including those with repeating decimals, as well as using this skill to solve mathematical and real-world problems. In grade 8, students will learn about irrational numbers as well as working to plot, order and compare rational and irrational numbers.

- When solving problems with numbers written in various forms, students must be able to convert between these forms to perform operations or make comparisons (*MTR.2.1*).
- Students should begin to develop charts, like the one below, that allow them to find patterns within the different forms of rational numbers. Students should have common fractions, decimals and percentages at their disposal in order to move to ones that are more difficult to determine.

Fraction	Mixed Number	Decimal	Percentage

- Students should have practice with and without the use of technology to rewrite rational numbers in different but equivalent forms.
- Students should work with simple problems to highlight how truncating repeated decimals may result in incorrect solutions.
 - For example, using the fractional value of $\frac{1}{3}$ may provide a more precise answer than using the truncated decimal of 0.33.

- $\frac{1}{3}(3000) = 1000$ versus $0.33(3000) = 990$ Repeating decimals can be written using a repeating bar, a truncated decimal form, fraction or may be rounded depending on the problem and the directions to the problem.
- Students should use reasonableness to determine if it is appropriate to use a specific equivalent form over another one when problem solving (*MTR.6.1*).

Common Misconceptions or Errors

- Students may not differentiate between terminating decimals, repeating decimals and rounded decimals, and they may not use them appropriately within the given contexts.
- Students may incorrectly truncate repeating decimals when problem solving.
- Students may incorrectly divide when the quotient is not a whole number.
 - For example, students may use the remainder of a problem as a decimal representation.

Strategies to Support Tiered Instruction

- Instruction includes the use of estimation to find the approximate decimal value of a fraction or mixed number before rewriting in decimal form to help with correct placement of the decimal point.
- Teacher provides opportunities for students to explore and discuss the differences between repeating and truncated decimals and the impact of truncating repeating decimals when solving problems.
 - For example, provide students with the equation $y = \frac{1}{3}x$ and have them create a table of values comparing using $\frac{1}{3}$, 0.3, 0.333 and 0.33333 as the constant of proportionality. Students can discuss the differences in y -values and importance of using exact values in some cases and approximate values in others.

x	$y = \frac{1}{3}x$	$y = 0.3x$	$y = 0.333x$	$y = 0.33333x$
1	$\frac{1}{3} = 0.\bar{3}$	0.3	0.333	0.33333
3	1	0.9	0.999	0.99999
5	$\frac{5}{3} = 1.\bar{6}$	1.5	1.665	1.66665
6	2	1.8	1.998	1.99998

- Instruction includes co-creating a graphic organizer to highlight the differences between terminating decimals, repeating decimals, and rounded decimals.

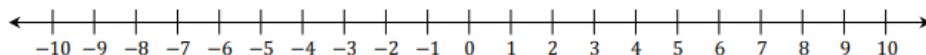
Instructional Tasks

Instructional Task 1 (MTR.2.1)

Convert each of the following to an equivalent form in order to compare their values.

$$\frac{1}{5} \quad -0.4 \quad 65\% \quad -2\frac{1}{3} \quad 5.75 \quad \frac{9}{7} \quad 123\% \quad 2.\bar{3}$$

Part A. Graph the numbers on a number line to determine increasing order.



Part B. Robin plotted her number line using all decimals, whereas Courtney plotted them using the original forms. Describe why both would be acceptable answers.

Instructional Task 2 (MTR.2.1, MTR.4.1, MTR.5.1)

Complete the table to identify equivalent forms of each number. Explain how you approached your solutions. *Prompting questions:* What patterns did you use? How did you start? Which values in the table are you most comfortable in starting with?

Fraction	Mixed Number (if applicable)	Decimal	Percentage
	$-4\frac{1}{3}$		
		0.125	
$\frac{29}{7}$			
$-\frac{8}{25}$			-32%
	$2\frac{7}{9}$	$2.\bar{7}$	

Instructional Items

Instructional Item 1

All of the students in first period were given a glue stick to help build their interactive notebook. Benny said he has already used $\frac{2}{3}$ of his glue while Juniper has used 70% of hers. Which student has the most glue remaining for their notebooks?

Instructional Item 2

Ishana manages a corner store and wishes to give a discount to her customers for the holiday. If she subtracts 0.15 of the cost of any item in the store, what percent should her sale sign promote?

Instructional Item 3

Write three equivalent forms for $5\frac{7}{8}$.